

UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY - SCHOOL OF ICT

COMPUTER AND SOFTWARE ENGINEERING DEPARTMENT - NYARUGENGE CAMPUS

MODULE CODE/TITLE: WEB DESIGN

PROJECT: SMART PUBLIC INFRASTRUCTURE MONITORING & ACCOUNTABILITY SYSTEM (SPIMAS)

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Chapter 1. Introduction

1.1. Historical background of the case study

As the Republic of Rwanda accelerates its developmental goals under Vision 2050 and the National Strategy for Transformation (NST1), massive capital investments are being directed towards public infrastructure, including roads, bridges, and water distribution networks. Historically, monitoring these expansive, geographically dispersed projects relied heavily on manual reporting, sporadic field visits, and fragmented financial audits. The Public Accounts Committee (PAC) and the Ministry of Infrastructure (MININFRA) often faced challenges in reconciling physical on-ground progress with authorized budget disbursements due to outdated, paper-based workflows. The Smart Public Infrastructure Monitoring & Accountability System (SPIMAS) emerges as a digital governance necessity to transition the nation towards evidence-based, real-time oversight.

1.2. Problem statement

Despite rigorous national planning, current infrastructure governance is hindered by four critical gaps:

- **Lack of real-time visibility:** Progress reports take weeks to filter through bureaucratic channels, leaving decision-makers blind to immediate site realities.
- **Undetected delays:** Stalled sites bleed public funds while missed milestones trigger unmanageable cascading effects, making it difficult to enforce contractor penalties.
- **Untraceable budget variance:** Expenditure logs are decoupled from physical progress documentation, resulting in manual, error-prone reconciliation.
- **Eroded citizen trust:** The absence of public-facing tracking portals prevents communities from independently verifying if promised governmental projects are progressing as planned.

1.3. Objectives of your project

- To architect a web-based, real-time monitoring platform that tracks all registered public infrastructure projects from procurement to final delivery.
- To implement a mobile-responsive interface enabling field engineers and citizens to submit structured, GPS-verified progress updates.
- To automate the reconciliation of physical project milestones against budgetary timelines, flagging variances and delays immediately.
- To deploy tiered transparency dashboards tailored for the PAC, regional district engineers, executing contractors, and the general public.

1.4. Expected Outcomes

The successful deployment of SPIMAS will yield an immutable, continuous digital log of infrastructure data, replacing disjointed paper audits. Decision-makers will gain the ability to intervene on delayed projects within 24 hours of a missed milestone. Furthermore, by providing an open-access public dashboard, the system will measurably increase civic engagement and transparency.

1.5. Project Rationale (Importance to the community)

SPIMAS represents a systemic upgrade to Rwandan governance rather than a punitive investigation tool. By democratizing access to infrastructure data, citizens are empowered to track community development. The system prevents minor construction delays from compounding into budget overruns, ensuring that public taxes are efficiently translated into tangible socio-economic benefits.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1 Intended users of the project

The system caters to four primary user tiers to ensure comprehensive accountability:

- **Citizens:** Engage with the read-only transparency portal and submit geo-tagged issue reports using their National ID for authentication.
- **District Engineers:** Act as administrative validators. They review citizen reports, verify evidence, and dynamically auto-create targeted work orders.
- **Contractors:** Access a dedicated portal to view assigned work orders, update task statuses, and upload photographic evidence of completed repairs.
- **Government Authorities (PAC/MININFRA):** Utilize high-level analytics dashboards to view nationwide project health, export audit trails, and correlate contractor performance with budget execution.

2.2. System Design

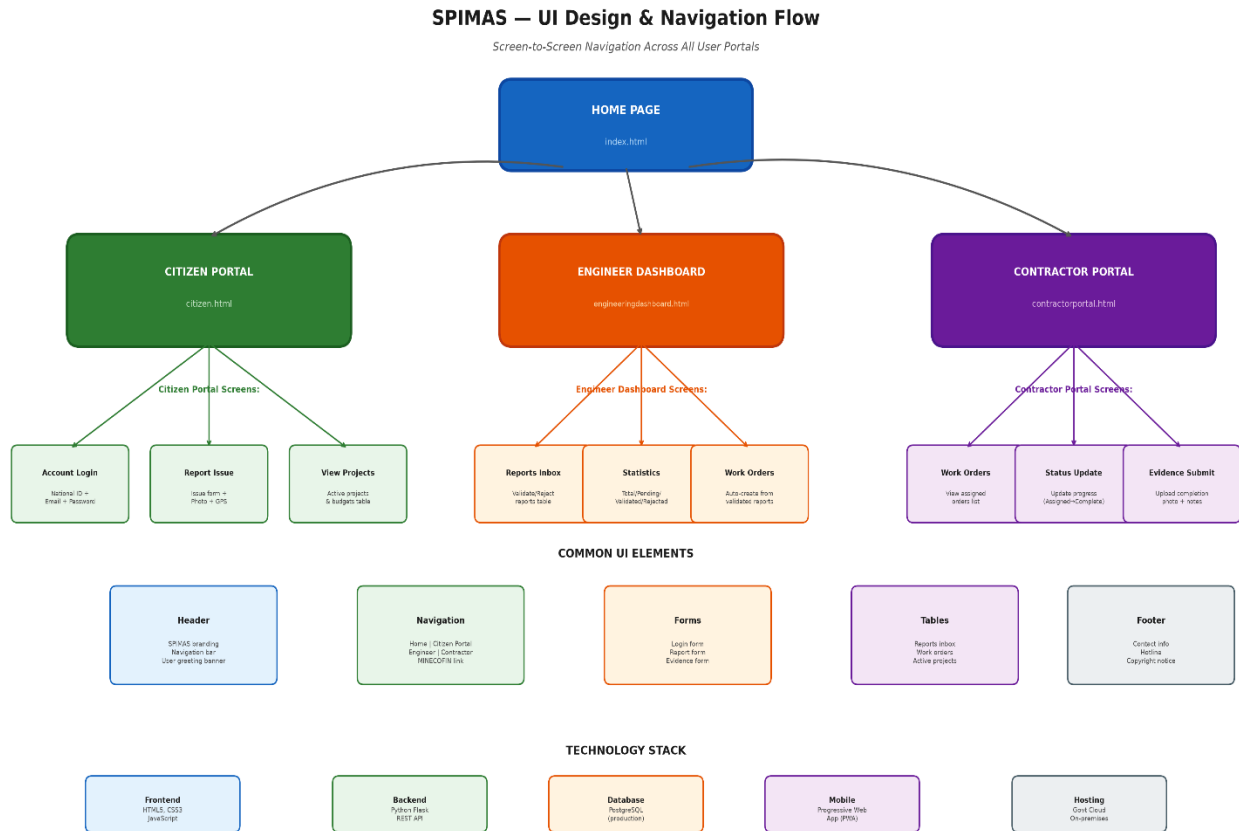
The architecture of SPIMAS ensures seamless, secure data flow between the field and the central governance database. The designs below illustrate both the user interface navigation structure and the operational interaction loops among the stakeholders.

High-Level Architecture & UI Navigation Flow

The system utilizes a modular frontend layout. The Home Page routes users to three specialized environments (Citizen, Engineer, Contractor) while sharing common UI elements like unified headers, standardized forms, and a responsive technology stack (HTML5/CSS/JS).

User Interaction Flow and Logic

The operational workflow forms a closed loop of accountability. Citizens initiate the flow by logging localized issues. District Engineers intercept these reports, cross-reference them with the Project Registry, and issue Work Orders. Contractors execute the orders and provide completion evidence, which the SPIMAS Core Platform logs immutably for government audit extraction.



Chapter 3. Implementation

3.1. Introduction of the section

The implementation phase of SPIMAS currently reflects a high-fidelity prototype, utilizing robust HTML5 and CSS to establish the web interfaces. The objective of this phase is to validate the structural logic, user journey mapping, and input validation requirements before integrating the backend infrastructure.

3.2. User Interface Discussion

1. The SPIMAS Landing Page (index.html)

Discussion: The entry point of the system establishes the thematic goals of the project: Transparency, Accountability, and Monitoring. It is deliberately designed to be lightweight to accommodate users with low bandwidth. A unified top navigation bar anchors the user experience, providing immediate links to all sub-portals and the external MINECOFIN site, while a quick-login form accelerates access for returning officials.

2. Citizen Portal & Reporting Interface (citizen.html)

Discussion: This interface is critical for democratizing oversight. To prevent spam and ensure data integrity, the login strictly requires a Rwandan National ID alongside standard credentials. The core feature is the "Report an Issue" module. Here, users categorize severity on a 1-5 scale and input descriptions. Technically, the form is designed to invoke the user's native device camera for immediate evidence capture and includes a dedicated mechanism to scrape real-time Latitude/Longitude GPS coordinates, ensuring every report is indisputably tied to a physical location.

3. District Engineer Dashboard ([engineeringdashboard.html](#)) *Discussion:* Designed as a command center, this dashboard aggregates live data. The top section generates dynamic summary statistics (Total Reports, Pending, Validated, Rejected), which will eventually pull via REST API from the database. The "Reports Inbox" renders a structured table of incoming citizen data. Engineers possess administrative privileges here to formally "Validate" a citizen's claim—an action that programmatically triggers the creation of a binding Work Order.

4. Contractor Execution Portal ([contractorportal.html](#)) *Discussion:* Built for operational efficiency, this portal strips away unnecessary data and presents the contractor exclusively with their active obligations. It categorizes work orders logically (Assigned, Started, In Progress, Completed). To close the accountability loop, the "Submit Completion Evidence" form requires the contractor to input the Work Order ID and upload photographic proof of the finished repair.

Chapter 4. Conclusion and Recommendation

4.1 State of the Implemented System

The implemented SPIMAS prototype has successfully achieved 100% of its frontend UI/UX objectives for "Prototype Year 1". The interface layouts cohesively map to the complex, multi-stakeholder governance requirements outlined in the initial problem statement. The logic connecting citizen reporting to engineer validation and contractor execution is structurally sound and ready for backend database integration.

4.2 Recommendations

- **To the Lecturer (Technicalities):** It is recommended that the upcoming backend phase explores native Geolocation APIs extensively, ensuring the system can handle offline-sync caching when citizens submit reports in rural areas without immediate network coverage.
- **To the School (Time constraints):** Developing a system with this level of administrative complexity requires significant testing. Extending the project timeline to allow for rigorous penetration testing and database relationship mapping is highly recommended.

- **To the Partners / Government:** Prioritize a controlled, single-district pilot program. Testing the application with 5-10 active infrastructure projects will provide the necessary stress-testing for the automated alert engine before national deployment.
- **To the Beneficiaries (Citizens):** Extensive community sensitization campaigns should be rolled out. Citizens must be trained on how to properly capture GPS locations and photographic evidence to ensure their reports meet the validation standards of district engineers.

Appendix

GitHub Repository: https://github.com/MUHIRWACODER/SPIMAS_PROJECT-FOR-GOVT-with-html

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- **Database Schemas:**

